

# Sai Vinaya Manaswi Seeda

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## PROFESSIONAL SUMMARY

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AI/ML Engineer with 4+ years of experience designing and deploying production-ready LLM applications, multi-agent systems, and RAG pipelines using Python and LangChain. Skilled in prompt engineering, guardrail design, NL-to-SQL architectures, vector database integration, and cloud deployment on GCP. Background spans industry and research, including deep learning, computer vision, and end-to-end AI pipeline development across diverse domains.

## EDUCATION

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### Alabama A&M University

*Master of Science in Computer and Information Sciences.*

Normal, AL

May 2025

### Jawaharlal Nehru Technological University Kakinada (JNTUK)

*Bachelor of Technology in Computer Science and Engineering.*

India

June 2022

## EXPERIENCE

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### AI Engineer

March 2026 – Present

*Novi Talent Group Inc*

*Coppell, TX*

- Architect and deploy multi-agent AI systems on Google Cloud Platform (GCP), engineering cloud infrastructure for LLM-powered REST APIs from initial deployment through ongoing operations.
- Design and implement MLOps workflows on GCP by integrating ChromaDB vector database hosting, automated CI/CD pipelines, and environment configuration management to ensure consistent, repeatable deployments across dev and production environments.
- Build a multi-agent AI system for sales analytics with specialized agents for SQL generation, query validation, data visualization, and business insight generation using Python and LangChain.
- Engineer a natural language to SQL (NL-to-SQL) pipeline with a SQL Validator Agent to convert natural language questions into executable SQLite queries, applying LLM guardrails and prompt engineering to detect and correct invalid syntax before execution.
- Develop a Retrieval-Augmented Generation (RAG) system using LangChain and ChromaDB with a full document ingestion pipeline covering PDF loading, text extraction, chunking, and embedding generation, using MMR retrieval and configurable top-k parameters.
- Design a ReAct-based AI agent integrating web search, word count validation, and file saving tools with dynamic reasoning and LLM-driven tool selection.
- Build an Insight Agent to generate business-level summaries from query outputs while minimizing hallucinations through structured prompt engineering and multi-layer validation.
- Develop an AI-powered travel planner with a 5-agent orchestrator architecture integrating 3 external APIs for flights, hotels, and weather data in a single request, with a form-based UI and conversational chatbot powered by Claude AI.
- Build conversational AI assistants using the Claude API and Ollama, enabling both cloud-based and fully offline, privacy-first AI interactions.
- Create automated data visualization pipelines using matplotlib with fallback mechanisms to ensure chart generation reliability when LLM-generated code fails.

### Mobility/Sustainability Systems Analyst

Aug 2025 – February 2026

*Alabama A&M University*

*Normal, AL*

- Designed a sustainability dashboard integrating transit, energy, and emissions data for real-time performance monitoring across university operations.
- Engineered automated reporting processes for mobility and sustainability KPIs, reducing manual reporting time by 30% through scripted data pipelines and template automation.
- Streamlined shuttle operations by analyzing route efficiency data, identifying scheduling inefficiencies that improved on-time performance and reduced conflicts.
- Implemented a digital signage solution for real-time transit updates and campus-wide sustainability announcements.
- Conducted data-driven evaluations of carbon footprint reduction initiatives, aligning outcomes with university sustainability targets.
- Presented analytical findings and recommendations to university leadership, influencing strategic decisions on future sustainability investments.

### Graduate Assistant

Aug 2023 – May 2025

*Alabama A&M University*

*Normal, AL*

- Applied machine learning and AI techniques to academic datasets, supporting data preprocessing, exploratory data analysis, and model evaluation workflows using Python.
- Processed and interpreted operational data from the Passio Bus Tracking System using AI-powered analytics tools, identifying performance patterns and generating actionable reports for transportation management.

- Built data analysis and visualization pipelines for transit and scheduling datasets, enabling data-driven decision-making for route planning and signage updates.
- Designed and maintained digital signage templates integrating real-time data feeds for dynamic content delivery of bus schedules, events, and emergency alerts.

**AI Research Intern**

May 2024 – Jul 2024

USDA

Maryland

- Processed and analyzed 192,000+ plant images to support AI and machine learning model development for agricultural research under the USDA-ARS AI-COE/SCINet program.
- Trained a YOLOv8-based computer vision model for object detection, achieving 90% accuracy in detecting and counting plant tillers for yield estimation.
- Built a deep learning model using convolutional neural networks (CNNs) for plant stress classification across three categories: healthy, stressed, and dead.
- Collected multispectral and thermal plant data using Phenospex and Planteye systems, extracting parameters including percent greenness, leaf temperature, and soil moisture for model training.
- Developed a labeled dataset using LabelMe for supervised machine learning tasks, annotating plant images to enable accurate model training and evaluation.
- Collaborated with USDA researchers to align model outputs with agronomic research objectives, adjusting training data and evaluation metrics based on domain feedback.

**Project Engineer**

May 2022 – May 2023

Wipro Ltd

Hyderabad, India

- Applied mainframe technology skills on live Citibank financial systems, contributing to enterprise-level support and software operations in a high-availability environment.
- Resolved support tickets across production systems to ensure timely solutions and uninterrupted operations, adhering to SLA requirements.
- Collaborated within global cross-functional teams to maintain service levels, coordinate deployments, and meet project delivery targets.

**Software Engineering Intern**

Jan 2022 – Mar 2022

Wipro Ltd

India

- Tested embedded systems code and integrated Java database connections for development and validation tasks.
- Prepared technical documentation, reports, and demo presentations for senior engineers to support project reviews and client demonstrations.

**SKILLS**

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**AI/ML Engineering:** Large Language Models (LLMs), Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, Guardrail Design, NL-to-SQL, Natural Language Processing (NLP), Computer Vision, Deep Learning, Convolutional Neural Networks (CNNs), Object Detection, Embeddings, Vector Databases, Semantic Search, MMR Retrieval, Machine Learning

**Programming Languages:** Python, SQL, Java, JavaScript, C, C++

**Frameworks and Libraries:** LangChain, TensorFlow, YOLOv8, ChromaDB, scikit-learn, matplotlib

**Cloud and MLOps:** Google Cloud Platform (GCP), CI/CD Pipelines, MLOps, REST API Development, Environment Configuration Management

**Tools and Platforms:** Claude API, Ollama, SQLite, LabelMe, Phenospex, Planteye, Microsoft Office (Word, PowerPoint, Excel)

**Other:** Model Training and Evaluation, Dataset Preparation, Data Visualization, Automated Reporting, KPI Analysis, Cross-functional Team Collaboration

**CERTIFICATIONS**

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- Microsoft Certified: Azure Data Fundamentals (DP-900)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Certified in Programming for Everybody (Python) - Coursera
- Certified in Introduction to Generative AI - Google Cloud